

GraphLIME on RDF knowledge graphs: explaining R-GCN entity classification in the vocabulary of the graph

XAI laboratory course — project report. Code, data pipeline and all numbers: the graphlime-rdf repository. Every table in this report is generated from results/*.jsonl by `python -m graphlime_rdf.cli tables` and injected by `cli readme`; no number is typed by hand.

1. Motivation

Relational Graph Convolutional Networks (R-GCN, Schlichtkrull et al., 2018) are the standard baseline for entity classification on RDF knowledge graphs. They are also opaque twice over: like any GNN their decision is buried in message passing, and — specific to the RDF setting — their usual input is a **one-hot entity id**, so even a perfect feature-attribution method could only say “entity #4211 mattered”, which explains nothing.

This project asks: *can we make the explanation speak RDF?* An explanation for “person X belongs to research group Y” should be a statement like “because X carries `swrc:publication` edges toward papers of that group” — predicates and terms of the knowledge graph, not tensor indices.

2. The design decision

The design below is not an optimisation but a precondition: **GraphLIME cannot be applied to an RDF graph as published.** Its entire mechanism operates on a node **feature matrix** — it collects the feature vectors of the target’s neighbours and asks, via kernels, which feature dimensions the model’s outputs depend on. The method was designed for citation networks, where every node carries a bag-of-words vector and every dimension is a word. In the RDF setting neither ingredient exists: entities carry **no feature vectors at all** — every piece of information is a typed edge toward another term — and the R-GCN’s customary one-hot input (an entity-id embedding) offers no dimension a human could read. With no feature matrix there is nothing for HSIC Lasso to select from, and with anonymous embedding dimensions any selection would be unreadable anyway. The explainer is not merely weakened; its input is undefined.

The solution we designed is to **manufacture the missing feature space from the graph itself**, in the graph’s own vocabulary (the methodological contribution of this project):

- **Space A — predicate counts.** Feature `out:<p>` counts the node’s outgoing `<p>` edges, `in:<p>` its incoming ones.
- **Space B — predicate-object pairs.** Feature `out:<p>=<o>` marks having `<p>` toward the specific term `<o>`, pruned below a support threshold.

Crucially, the R-GCN consumes space A directly as its node feature matrix, in place of the one-hot input. Model and explainer therefore share one vocabulary: any feature the explainer scores is a readable statement about the graph (“has publication edges”, “is author of ...”), and fidelity — otherwise undefined, since the explainer’s features would not be what the model consumes — becomes well-defined as masking feature columns of the actual model input.

3. Method

3.1 The model: R-GCN

The R-GCN (Schlichtkrull et al., 2018) extends graph convolution to multi-relational graphs by giving every relation type its own transformation. The layer-wise propagation rule for node i is

$$h_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} W_r^{(l)} h_j^{(l)} + W_0^{(l)} h_i^{(l)} \right),$$

where $\mathcal{N}_i^r$ is the set of neighbours reached from i through relation r , $c_{i,r} = |\mathcal{N}_i^r|$ is a per-relation normalisation constant, $W_r^{(l)}$ is the weight matrix specific to relation r , and $W_0^{(l)}$ is a self-loop transformation. The per-relation weights are the point: in a knowledge graph the meaning lives in the edge *types*, and a plain GCN would collapse, say, author and employs into an undifferentiated “neighbour”. Since one full matrix per relation is prohibitive (45 relations on AIFB), each W_r is expressed by **basis decomposition**,

$$W_r^{(l)} = \sum_{b=1}^B a_{rb}^{(l)} V_b^{(l)},$$

a learned linear combination of B shared basis matrices — parameter sharing that doubles as regularisation. Two such layers followed by a softmax, trained with cross-entropy on the few labelled entities, constitute the classifier we explain.

3.2 HSIC: measuring non-linear dependence

The Hilbert–Schmidt Independence Criterion (Gretton et al., 2005) is a kernel measure of statistical dependence between two variables. Given n paired samples, kernel matrices K and L over the two variables, and the centering matrix $H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$, the empirical estimator is

$$\widehat{\text{HSIC}}(K, L) = \frac{\text{tr}(KHLH)}{(n-1)^2}.$$

For characteristic kernels (we use RBF kernels with median-heuristic bandwidth) HSIC vanishes exactly when the variables are independent — and, unlike correlation, it detects **non-linear** dependence. The canonical example: $y = x^2$ with x symmetric around zero has Pearson correlation ≈ 0 but clearly positive HSIC. This single example justifies the whole construction — a GNN’s output is a non-linear function of its input, so a linear surrogate would measure the wrong thing. Our property tests assert this behaviour explicitly.

3.3 HSIC Lasso and its three-term reading

HSIC Lasso (Yamada et al., 2014) turns HSIC into a feature selector. For features $x_1, \dots, x_d$ and output y , build the centered, Frobenius-normalised kernel matrices $\bar{K}_1, \dots, \bar{K}_d$ and $\bar{L}$, and solve the non-negative Lasso

$$\min_{\beta \geq 0} \frac{1}{2} \left\| \bar{L} - \sum_{k=1}^d \beta_k \bar{K}_k \right\|_F^2 + \rho \|\beta\|_1.$$

Expanding the squared Frobenius norm exposes the **explainable three-term form** (up to the constant $\frac{1}{2} \|\bar{L}\|_F^2$):

$$\min_{\beta \geq 0} \underbrace{- \sum_k \beta_k \text{HSIC}(x_k, y)}_{\text{relevance}} + \underbrace{\frac{1}{2} \sum_{k,l} \beta_k \beta_l \text{HSIC}(x_k, x_l)}_{\text{redundancy}} + \underbrace{\rho \|\beta\|_1}_{\text{sparsity}}.$$

In words: the first term rewards features the output depends on; the second penalises features that are redundant with one another (a duplicated feature splits its weight instead of doubling it — asserted by our tests); the third forces most weights to zero. The resulting $\beta_k \geq 0$ is feature k 's importance.

3.4 GraphLIME and why it must be modified for RDF

GraphLIME (Huang et al., 2020) departs from LIME in that it perturbs nothing: to explain node v , the “samples” are the **real nodes of v 's h -hop neighbourhood** (here $h = 2$). X collects their feature vectors, Y the model's predicted probability vectors; feeding both to HSIC Lasso yields β , a sparse non-negative importance over *feature dimensions*. Two consequences shape this project. First, **GraphLIME explains features, not edges** — so on a graph whose nodes have no features the method has no input, which is precisely the RDF situation of §2. Second, its samples are a local neighbourhood: a small neighbourhood means few samples and a meaningless HSIC estimate, so the pipeline refuses to explain nodes below a minimum neighbourhood size rather than emit noise.

Our modification is the feature space of §2, in two variants: **method A** (predicate counts — one feature per predicate and direction; coarse, dense, stable, ≈ 90 columns on AIFB) and **method B** (predicate-object pairs — expressive, e.g. “has member toward group Y ”, but high-dimensional and sparse, which starves the kernel estimates; hence the support-threshold pruning). Comparing A against B is itself one of the reported experiments (§5).

3.5 The pipeline

End to end: (1) parse the raw RDF into a deterministic multigraph; (2) build the feature vocabulary and matrix X (space A, and B for the comparison); (3) train the R-GCN *on X* — not on one-hot ids — so model and explainer share the vocabulary; (4) for every test entity, extract the 2-hop neighbourhood, run HSIC Lasso on the neighbourhood's features against the model's predicted probabilities, and map the non-zero β back to predicate names; (5) evaluate with the metrics of §5, gated by the synthetic ground-truth check of §4. Both HSIC and the solver are implemented from scratch (explain/hsic.py, explain/hsic_lasso.py) with hypothesis property tests written

before the implementation: symmetry, non-negativity, $nHSIC \in [0,1]$, joint-permutation invariance, weight-splitting for duplicated features, and sparsity monotone in the regularisation ρ .

Full derivations as implemented: `docs/method.md`.

Baseline. GNNExplainer (Ying et al., 2019), as shipped by PyG. Its edge masks are structurally incompatible with RGCNConv (one propagate per relation on edge subsets — the full-graph mask can never align; ADR-007), so the baseline uses GNNExplainer’s *attribute mask* over the same feature matrix: it scores exactly the features GraphLIME scores, making rankings directly comparable at the predicate level.

4. Experimental setup

Datasets. AIFB (8,285 nodes, 29,043 triples, 45 predicates, 4 classes — research-group affiliation) and MUTAG (23,644 nodes, 74,227 triples, 23 predicates, 2 classes — mutagenicity), the standard R-GCN benchmark dumps, parsed directly from the raw RDF with fully deterministic indexing (blank nodes canonicalised; loader byte-identical across processes and PYTHONHASHSEED). The label-revealing predicates (employs/affiliation for AIFB, isMutagenic for MUTAG) are already stripped from the benchmark dumps; an explicit blocklist **asserts** their absence at every load.

Model. Two-layer RGCNConv with basis decomposition (30 bases, hidden 32, dropout 0), full-batch Adam (lr 0.01, 200 epochs), predicate-count features in both directions, 5 seeds, deterministic kernels. Chosen from a small fixed grid (ADR-006); per-seed accuracies live in `runs/final/` and in the tables below. For reference, the R-GCN paper reports 95.8% (AIFB) and 73.2% (MUTAG) with one-hot inputs.

Explanations. GraphLIME with 2-hop neighbourhoods (deterministically subsampled above 200 nodes), RBF kernels with median-heuristic bandwidth, $\rho = 0.1$. Neighbourhoods under 10 nodes are refused and counted. All test nodes of each dataset are explained.

Correctness gate. Before any number below was produced, the synthetic ground-truth gate (M6) had to pass: on a generated graph where class 1 $\iff$ the entity has predicate p_7 , GraphLIME ranked out:syn:p7 **first** for $\geq 95\%$ of explained nodes at 0% noise, and kept it top-3 under two distractor predicates 90%-correlated with the target. It passed on the first run (just synthetic reproduces it). The fidelity+ > random-k requirement is likewise asserted on the synthetic graph only; on real data the numbers are reported, not asserted.

5. Results

5.1 The evaluation metrics

There is no ground-truth explanation on real data, so the pipeline is judged by a set of complementary quantities, defined below. Throughout, V is the set of explained test entities, $v \in V$ one of them, x_v its feature vector, $\hat{y}$ the class the model predicts for it, $f_{\hat{y}}(\cdot)$ the model’s predicted probability for that class, and S_k the set of the k features with the largest β in v ’s explanation.

Classification accuracy. The fraction of held-out test entities whose predicted class is correct, reported as mean $\pm$ standard deviation over the 5 training seeds. It is not a measure of explanation quality; it establishes that the interpretable feature space is a viable model input at all, since an explanation of an untrained model would be worthless.

Fidelity+ (necessity). Zero the top- k feature columns of the model’s input and re-run the forward pass:

$$\text{fid}^+ = \frac{1}{|V|} \sum_{v \in V} \left[f_{\hat{y}}(x_v) - f_{\hat{y}}(x_v^{\setminus S_k}) \right].$$

If the selected features are the ones the model relies on, removing them should make the predicted-class probability fall, so **higher is better**.

Fidelity– (sufficiency). The complementary operation: keep *only* the top- k columns and zero everything else,

$$\text{fid}^- = \frac{1}{|V|} \sum_{v \in V} \left[f_{\hat{y}}(x_v) - f_{\hat{y}}(x_v^{S_k}) \right].$$

If the selected features are sufficient on their own, the prediction should survive, so **lower is better**.

Random- k control. Neither number is meaningful in isolation: masking any k columns perturbs the prediction somewhat. Both metrics are therefore recomputed with a random k -subset in place of S_k , and the quantity that carries the evidence is the **gap** — fid^+ above random^+ , fid^- below random^- . The sweep $k \in \{1, 2, 5, 10\}$ doubles as the sparsity axis: a good explanation should already win at small k , since the explanations themselves typically carry fewer than ten non-zero β .

Stability (Jaccard@5). Agreement between two top- k feature sets,

$$J(S, S') = \frac{|S \cap S'|}{|S \cup S'|} \in [0, 1],$$

averaged over the explained nodes with $k = 5$. It is computed along two axes: across **training seeds** (models of equal accuracy trained from different initialisations) and across **hops** (the same model explained with 1-, 2- and 3-hop neighbourhoods). An explanation method that changes its answer when nothing meaningful changed is not trustworthy, so higher is better; the hop axis additionally quantifies how much the neighbourhood radius — GraphLIME’s sample source — dictates the result.

Agreement (Jaccard@5 over predicates). The same Jaccard index, but with each feature first collapsed to its predicate URI so that the two feature spaces and the baseline become comparable. It is used twice: **space A vs space B**, measuring how much the choice of interpretable vocabulary changes the conclusion, and **GraphLIME vs GNNExplainer**, an external cross-check. Neither is a correctness measure — there is no reference ranking — so it is read as a sensitivity analysis rather than a score.

Refusal rate. The fraction of test entities whose neighbourhood falls below `min_neighborhood = 10` and for which the pipeline declines to produce an explanation. Reporting it is part of the method: a refusal is a finding about applicability, not a failure.

Classification accuracy

dataset	test accuracy (mean $\pm$ std, 5 seeds)	per-seed	best seed	best accuracy	checkpoint sha256
AIFB	0.8833 ± 0.0534	0.9167, 0.9444, 0.8056, 0.8889, 0.8611	1	0.9444	8f69360a52b0
MUTAG	0.7559 ± 0.0305	0.7794, 0.7206, 0.7353, 0.7500, 0.7941	4	0.7941	20de0d5614ec

Explanation fidelity (column masking, random-k control)

dataset	k	fidelity+	random+	fidelity—	random—	nodes
AIFB	1	0.0176	0.0129	0.7145	0.6767	36
AIFB	2	0.0162	0.0066	0.7386	0.7556	36
AIFB	5	0.1079	0.0520	0.7160	0.6085	36
AIFB	10	0.1829	0.0735	0.6338	0.6184	36
MUTAG	1	0.0789	-0.0013	0.5397	0.3151	68
MUTAG	2	0.1224	0.0205	0.3068	0.2744	68
MUTAG	5	0.1627	0.0475	0.1987	0.3352	68
MUTAG	10	0.0999	0.1592	0.2299	0.3462	68

Explanation stability (Jaccard@5)

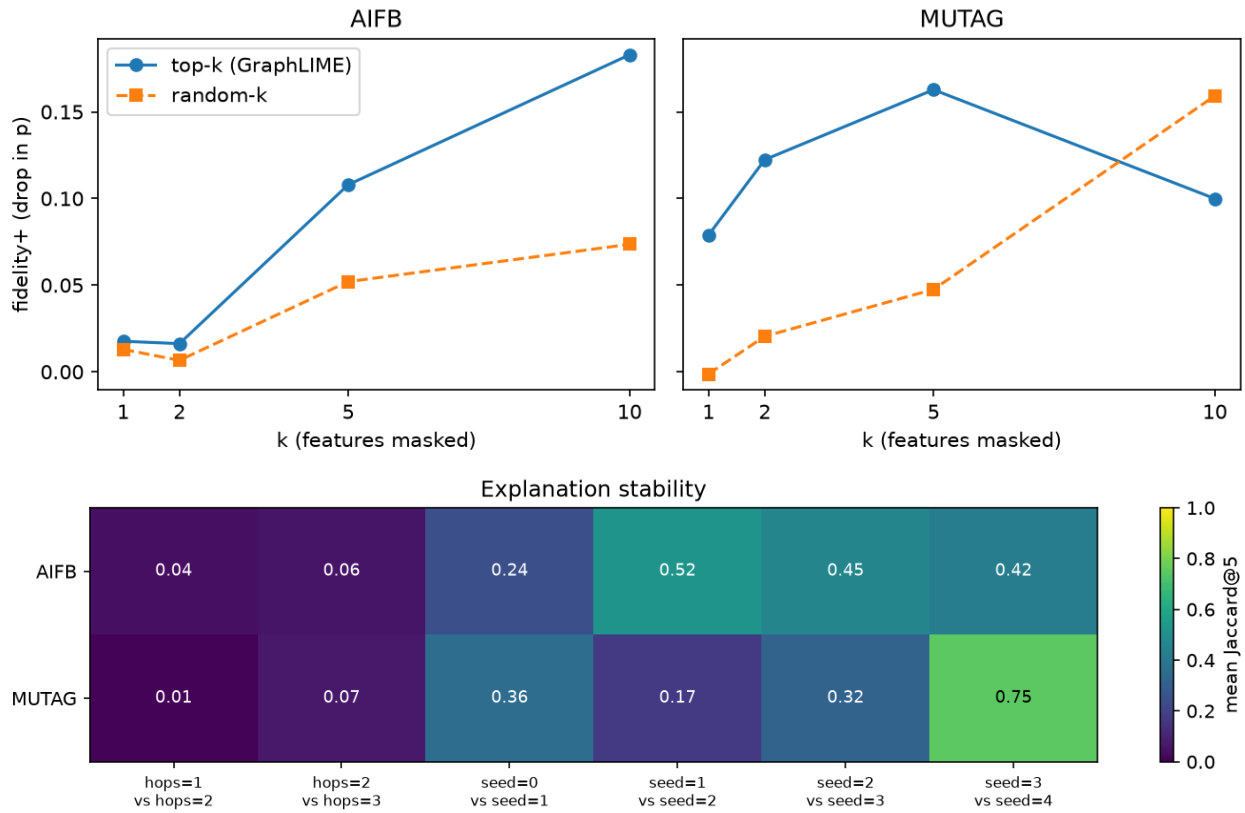
dataset	comparison	pair	mean Jaccard@5	pairs
AIFB	hops	hops=1 vs hops=2	0.0413	21
AIFB	hops	hops=2 vs hops=3	0.0585	36
AIFB	seeds	seed=0 vs seed=1	0.2443	36
AIFB	seeds	seed=1 vs seed=2	0.5162	36
AIFB	seeds	seed=2 vs seed=3	0.4547	36
AIFB	seeds	seed=3 vs seed=4	0.4239	36
MUTAG	hops	hops=1 vs hops=2	0.0103	68
MUTAG	hops	hops=2 vs hops=3	0.0696	68
MUTAG	seeds	seed=0 vs seed=1	0.3593	68
MUTAG	seeds	seed=1 vs seed=2	0.1709	68
MUTAG	seeds	seed=2 vs seed=3	0.3179	68
MUTAG	seeds	seed=3 vs seed=4	0.7475	68

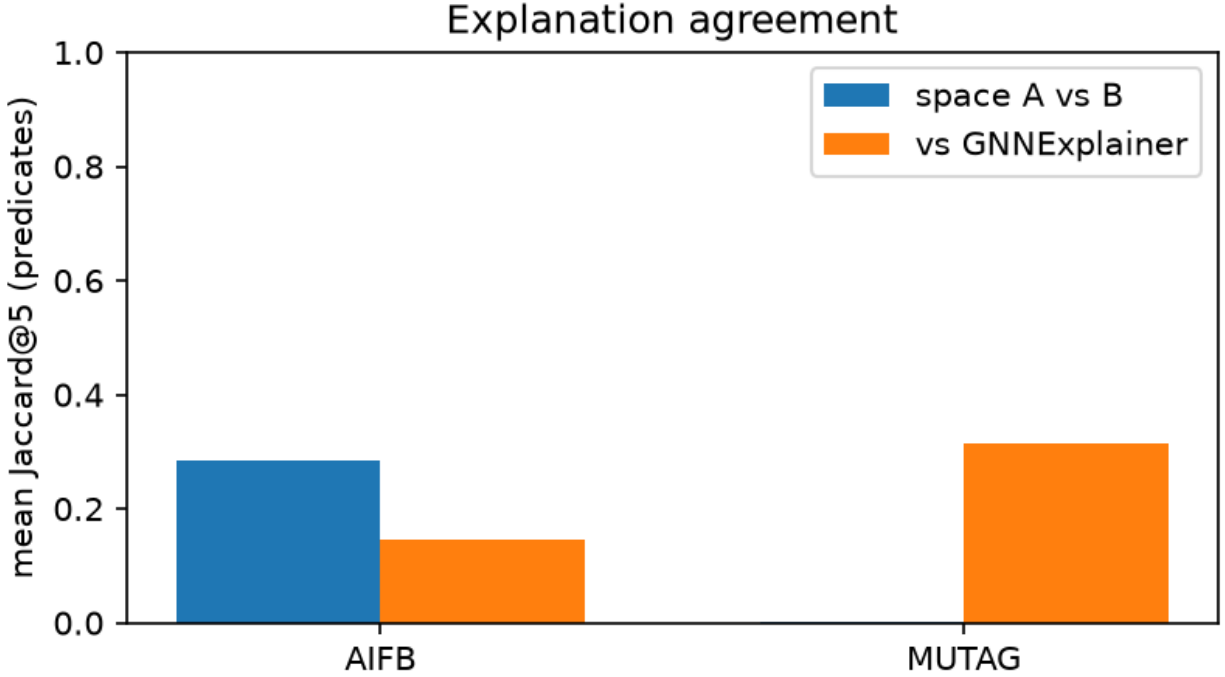
Feature-space sensitivity and baseline agreement

dataset	comparison	mean Jaccard@5 (predicates)	nodes
AIFB	GraphLIME: space A vs space B	0.2843	36
AIFB	GraphLIME vs GNNExplainer	0.1456	36
MUTAG	GraphLIME: space A vs space B	0.0029	68
MUTAG	GraphLIME vs GNNExplainer	0.3153	68

Refused explanations

dataset	explained	refused	refusal rate	reasons
AIFB	36	0	0.0000	—
MUTAG	68	0	0.0000	—





5.2 Reading the results, table and figure by figure

Classification accuracy (table). The interpretable feature space is not a handicap on MUTAG — the 5-seed mean, 0.7559, *exceeds* the paper’s one-hot R-GCN (0.732) — while AIFB lands several points below its 0.958. The gap is informative rather than disappointing: AIFB’s classes hinge on *which* specific entity a person is connected to, exactly the information that per-entity embeddings encode for free and that predicate counts deliberately discard in exchange for readability. The cost of interpretability is thus visible and dataset-dependent, and it is paid in the input representation, not in the explainer.

Fidelity+ vs k against the random-k control (`fidelity_curve.png`, plus the fidelity table). For $k \leq 5$ the GraphLIME curve sits clearly above the random- k control on both datasets — on MUTAG at $k = 1$, 0.079 against -0.001 , and on AIFB at $k = 5$, 0.108 against 0.052 — so masking the selected columns damages the prediction more than masking arbitrary ones: the explanations point at features the model actually uses. At $k = 10$ on MUTAG the comparison inverts (0.100 against 0.159). This is a property of the sweep, not a failure of the method: explanations there typically have fewer than ten non-zero β , so the top-10 set pads itself with irrelevant columns, while random-10 already covers roughly a fifth of MUTAG’s 46 columns and is therefore a strong perturbation in its own right. The sufficiency side is read from the same table: fidelity— stays high on AIFB at every k (≈ 0.7), meaning no five features *suffice* — the model spreads its evidence across the whole neighbourhood feature profile — whereas on MUTAG keeping only the top-5 preserves the prediction markedly better than keeping random-5 (0.199 against 0.335).

Stability heatmap across seeds and hops (`stability_heatmap.png`). The two axes of the heatmap tell opposite stories. Across **training seeds**, explanations agree moderately, with mean Jaccard@5 between 0.17 and 0.75: models of comparable accuracy reach their decisions through partially different feature use, which is exactly why the per-seed comparison is reported instead of a single-seed number. Across **hops**, agreement is near zero (0.01–0.07). The reason is structural: the 1-hop and 2-

hop neighbourhoods of an entity are different node populations, and since GraphLIME’s samples *are* the neighbourhood, the local HSIC estimate follows the sample rather than the target alone. The neighbourhood radius is therefore the single most consequential explanation hyperparameter in this setting, and reporting it as such is one of this project’s honest negative findings.

Feature-space (A vs B) and baseline agreement (`agreement_comparison.png`). Both comparisons are made at the predicate level. AIFB shows moderate A-vs-B overlap (0.28), MUTAG essentially none (0.003): the pair-level space shifts weight onto specific `rdf:type` objects — individual atom types — which space A cannot express at all, so the two vocabularies describe genuinely different things on that dataset. Agreement with GNNExplainer’s attribute mask is limited on both (0.15 on AIFB, 0.32 on MUTAG). This is expected rather than alarming: kernel-based dependence selection and a gradient-learned soft mask answer related but distinct questions, and with no reference ranking available the disagreement is itself the reported result.

Refused explanations (table): none. With 2-hop neighbourhoods every test entity of both datasets clears `min_neighborhood = 10`, giving a refusal rate of 0.0000 on both. The refusal machinery is nonetheless exercised by the test suite and by smaller-radius runs; here it is reported empty, which is the correct reading of “the guard was never needed at this radius” rather than “the guard is absent”.

6. Qualitative examples

The executed notebook (`notebooks/qualitative.ipynb`, rendered as `qualitative.html`) loads the committed checkpoints and prints sentence-form explanations for test entities of both datasets, e.g. a person *correctly predicted* into a research group *because of* publication and author edges toward the group’s papers, or a MUTAG bond classified through its atom-type context. Refusals are printed inline, not skipped.

7. Limitations

The β weights are dependence, not causation. HSIC Lasso ranks features by how strongly the model’s outputs *co-vary* with them over the local sample, after discounting redundancy among the features themselves. A high β_k therefore licenses the statement “within this neighbourhood, the model’s prediction depends on predicate k ”, not “removing predicate k would change the entity’s true class”. The account is additionally **local**: it is fitted on one neighbourhood and does not generalise into a global statement about the model, which is by design — GraphLIME is a local surrogate — but must be remembered when reading a single explanation as though it described the classifier as a whole.

Fidelity is measured on columns, not on structure. Masking a feature column removes the *evidence that the entity has a predicate*, but leaves the graph’s edges intact, so message passing still propagates along them. This is the honest consequence of the design decision in §2: model and explainer share the feature vocabulary, which is what made fidelity definable at all, but the perturbation lives in the feature matrix rather than in the RDF triples. A structurally faithful alternative — deleting the corresponding triples and re-running the forward pass — would respect RDF semantics more closely at a substantially higher cost, and was deliberately not run here; the fidelity numbers should be read as necessity and sufficiency of feature *columns*, graph-wide.

Neighbourhood handling introduces two distortions at the extremes. Above 200 nodes the neighbourhood is subsampled, trading a small amount of estimator variance for tractability; the subsam-

ple is deterministic per node and always retains the target, so results are reproducible, but very large neighbourhoods are described by a sample rather than in full. Below 10 nodes the pipeline refuses outright, because HSIC estimated from a handful of samples is noise. Between those bounds, §5.2 shows that the radius itself dominates the outcome — the near-zero cross-hop Jaccard means that “the explanation of entity v ” is only well-defined once the radius is fixed and stated.

The evidence base is two small benchmarks. AIFB and MUTAG have 176 and 340 labelled entities respectively, which puts every mean in §5 on a few tens of test nodes and makes the seed-to-seed spread in both accuracy and stability substantial. Conclusions about the interpretable feature space are therefore claimed for RDF entity classification at this scale and not beyond it; in particular, the A-vs-B comparison reflects the vocabulary of these two ontologies, and space B’s behaviour is contingent on a support threshold whose sweep was not performed.

The baseline comparison is not like-for-like. GNNExplainer natively optimises a soft mask over *edges*, but its edge masks are structurally incompatible with RGCNConv, which runs one propagate per relation on edge subsets so that a full-graph mask can never align (ADR-007). The baseline therefore uses GNNExplainer’s *attribute* mask over the same feature matrix. This makes the two rankings directly comparable, at the price of comparing the two methods on feature attribution only — the edge-level explanation that GNNExplainer would normally produce is outside the comparison, and the agreement figures in §5.2 must be read accordingly.

References

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